



The physiological effects of induced prepartum subclinical hypocalcemia in multiparous Holstein cows

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ABSTRACT

Subclinical hypocalcemia (SCH) is prevalent during the periparturient period in dairy cows due to increased calcium (Ca) demand for milk and colostrum synthesis. Transient SCH has been shown to activate a negative feedback loop, thereby improving Ca homeostasis. Here, we investigate the impact of induced SCH during the prepartum period on Ca homeostasis pre- and postpartum, and milk production. Thirty multiparous Holstein cows were enrolled 21 d before the expected due date, in a randomized complete block design, and were assigned to receive either a 6-h continuous intravenous infusion of saline solution (CON) or 5% egtazic acid (EGTA) solution ($n = 15/\text{treatment}$) for a minimum of 3 d and a maximum of 14 d until parturition. Blood samples were collected daily from enrollment to the onset of infusion, and from 1 to 49 DIM (d 1 to 49). During infusions, blood samples were collected immediately before infusion, hourly during infusion, and 1 h post-infusion termination. Milk samples were collected weekly during the initial 3 wk of lactation, and milk weights were recorded daily from d 0 to 49. Results demonstrated that prepartum EGTA treatment led to decreased ionized Ca (iCa) and total Ca (tCa) concentrations compared with the control group during the infusion. Postpartum, EGTA-treated cows exhibited higher iCa on d 0 h 12 (12 h postpartum), d 1, and d 2, and higher tCa concentrations on d 0, d 0 h 12, d 1, d 2, d 3, and d 5. Additionally, EGTA-treated cows developed resistance to induced SCH, as indicated by an increased infusion rate (IR) on d –3, d –2, and d –1 compared with the first day of infusion (622.68 ± 40.19 mL/h). We observed a treatment $\times$ day $\times$ time interaction in parathyroid hormone (PTH) concentrations, with

EGTA-treated cows having higher concentrations at d –2 h 0 and d –1 h 0 compared with CON-treated cows. During the postpartum period, EGTA-treated cows had increased PTH concentrations compared with CON-treated cows. Additionally, EGTA-treated cows tended to have higher serotonin concentrations during the prepartum period compared with CON-treated cows. A treatment $\times$ day interaction on milk production was observed, with EGTA-treated cows having decreased milk yield on d 16, and d 18 to 24; however, there were no significant differences in milk production over the entire lactation period. These findings indicate that prepartum SCH induction primes Ca homeostatic mechanisms, potentially mitigating postpartum SCH.

Key words: calcium, hypocalcemia, EGTA

INTRODUCTION

The periparturient period is generally defined as the 3 wk pre- to postpartum, and it is a pivotal period for dairy cows due to an increased calcium (Ca) requirement for colostrum and milk production. Despite low clinical hypocalcemia (CH) incidence in US dairy herds, subclinical hypocalcemia (SCH) prevalence can occur in up to 50%, contingent upon lactation number (Reinhardt et al., 2011). As defined by the *Merck Veterinary Manual* (Merck & Co. Inc., 2024), SCH manifests as a blood total Ca (tCa) concentration below 2.0 mM. Nonetheless, caution is warranted in singular assessments within 48 h postpartum. Recent research highlights the potential benefits of transient SCH on the first DIM, followed by normocalcemia at 4 DIM, which promotes dairy cow health and productivity (McArt and Neves, 2020; Seely et al., 2021).

Transient SCH is crucial for activating the endocrine negative feedback loop(s) that regulate Ca, thereby prompting the parathyroid gland to increase its secretion of parathyroid hormone (PTH; Horst et al., 1994,

Received August 5, 2025.

Accepted December 23, 2025.

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

2005; Goff, 2008). The action of PTH occurs through the augmentation of Ca resorption in bones and enhancing renal Ca reabsorption, along with promoting the conversion of 25(OH)D₃ isoform to the active form of vitamin D (1,25(OH)₂D₃) (Potts, 2005; Goff, 2008). Furthermore, 1,25(OH)₂D₃ enhances Ca resorption in bones, Ca reabsorption in kidneys, and facilitates Ca absorption in the intestines (Bronner, 1987). Nonetheless, the demands of lactation necessitate a substantial Ca supply beyond the means of stimulation of the conventional Ca-PTH-1,25(OH)₂D₃ axis.

In addition to the traditional Ca feedback loop, serotonin (5-HT) and parathyroid hormone-related protein (PTHrP) play critical roles in Ca regulation. Studies have shown that cows infused with 5-hydroxy-L-tryptophan (5-HTP), the immediate precursor of 5-HT, exhibited a decrease in circulating Ca concentrations while experiencing increased concentrations of blood 5-HT and milk Ca (Laporta et al., 2015; Connelly et al., 2021). The authors suggested that 5-HTP infusion elevates circulating 5-HT, which subsequently drives Ca from the circulation to the mammary gland (MG), inducing transient hypocalcemia and increasing milk Ca concentrations (Laporta et al., 2015; Connelly et al., 2021). Furthermore, 5-HTP infusion in mid-lactation cows was found to increase *PTHrP* and calcium-sensing receptor (*CASR*) mRNA expression in the MG, indicating that 5-HT mediates an additional mechanism that modulates Ca homeostasis (Connelly et al., 2021). Lactating cows exhibit elevated circulating and MG 5-HT concentrations, which are known to stimulate the secretion of PTHrP (Matsuda et al., 2004; Collier et al., 2012). It has been demonstrated that lactating mammals synthesize and secrete PTHrP, which acts through the same G-protein coupled receptor as PTH and elicits similar actions, thereby complementing the regulatory role of PTH (VanHouten et al., 2003; Wysolmerski, 2012). These findings underscore the complex interplay between 5-HT, PTHrP, and traditional Ca regulatory mechanisms, highlighting the multifaceted nature of Ca homeostasis in lactating dairy cows.

In a recent experiment, egtazic acid (EGTA), a Ca chelator, was administered to nonpregnant, nonlactating, and early-lactation Holstein cows to induce SCH for a 24-h period (Connelly et al., 2022). Interestingly, early-lactation cows exhibited greater resistance to EGTA-induced SCH compared with their nonpregnant, nonlactating counterparts, necessitating a larger volume of EGTA to induce SCH (Connelly et al., 2022). Conversely, nonpregnant, nonlactating cows required a reduced and more constant EGTA infusion rate (IR) to maintain SCH (Connelly et al., 2022). Remarkably, upon cessation of the EGTA infusion, early-lactation cows demonstrated a rapid increase in circulating Ca concentrations, surpassing dry cows and sustaining higher blood Ca concentra-

tion for the subsequent 48 h (Connelly et al., 2022). In another investigation, cows were treated with either Ca gluconate or dextrose for 24 h immediately postpartum to prevent the typical decline in blood Ca concentrations after parturition (Connelly et al., 2024). The infusion of Ca gluconate resulted in a robust decrease in Ca concentration after termination of the infusion, which was maintained at lower concentrations for 24 h post-infusion (Connelly et al., 2024). These studies underscore the importance of transient SCH at parturition in activating Ca homeostatic mechanisms.

A previous epidemiological study reported a positive association between reduced plasma Ca concentration on 1 DIM and an increase in milk production (Neves et al., 2018). More recently, Seely et al. (2021) characterized SCH dynamics with DMI and milk production and observed that cows with normocalcemia (high Ca concentration on both 1 and 4 DIM) and those with transient SCH (low Ca concentration on 1 DIM with recovery by 4 DIM) had higher postpartum DMI, supporting effective Ca homeostasis and, consequently, had higher milk production (Seely et al., 2021). Conversely, cows with delayed (high Ca concentration on 1 DIM followed by a decrease on 4 DIM) and prolonged (low Ca concentration on both 1 and 4 DIM) SCH demonstrated compromised Ca homeostasis and produced less milk (Seely et al., 2021).

Therefore, the principal aim of this study was to assess the impact of prepartum-induced SCH on postpartum Ca homeostasis. A secondary aim was to determine the effects of prepartum SCH on postpartum DMI and milk production. We hypothesized that cows experiencing prepartum SCH induction would result in primed Ca homeostatic mechanisms before parturition, thereby mitigating the incidence of postpartum SCH.

MATERIALS AND METHODS

All animal and experimental procedures were approved by the Animal Care and Use Committee of the College of Agriculture and Life Sciences, University of Wisconsin–Madison, protocol number A006319-A03, and strictly followed.

Cows and Experimental Design

The experiment was conducted at the USDA Dairy Forage Research Center, Agriculture Research Service (Prairie du Sac, WI). The experiment was conducted from November 2020 to April 2021. Cows were housed in a cross-ventilated tiestall facility with ad libitum access to water. Herd TMR was provided once daily to dry cows (neutral DCAD) and lactating cows (Tables 1 and 2), with orts weighed the following day to calculate daily DMI for individual cows. Lactating cows were

Table 1. Dietary ingredient composition of dry and lactating cow diets

Ingredient (% DM)	Diet	
	Dry	Lactating
Wheat straw	28.03	—
Dry cow mineral and vitamin mix ¹	1.59	—
Urea	0.37	—
Roasted soybean	6.23	7.49
Rumen-protected choline ²	0.37	—
High-moisture shelled corn	9.34	21.29
Grass silage	0.03	—
Corn silage	40.48	31.61
Canola meal	13.55	10.48
Alfafa haylage	—	26.62
Lactating cow mineral and vitamin mix ³	—	2.50

¹Dry cow mineral and vitamin mix. Contents: 51.66% calcium sulfate; 16.65% magnesium oxide; 11.1% magnesium sulfate; 10.15% sodium chloride; 4.45% monocalcium phosphate; 2.20% bovine trace; 1.5% mineral oil; 0.43% Rumensin 90 (Elanco Animal Health); 518,000 IU/kg vitamin A; 137,900 IU/kg vitamin D; 6,900 IU/kg vitamin E.

²ReaShure (Balchem).

³Lactating cow mineral and vitamin mix. Contents: 16% calcium carbonate; 14% sodium sesquicarbonate; 10.25% sodium chloride; 5.85% magnesium oxide; 0.68% potassium magnesium sulfate (Dynamate, The Mosaic Company); 0.046% Rumensin 90 (Elanco Animal Health); 36.4 mg/kg cobalt carbonate; 442 mg/kg copper chloride (IntelliBond C^{II}, Micronutrients); 51.3 mg/kg calcium iodate; 2,235 mg/kg manganese sulfate; 13.2 mg/kg selenium selenite; 2,375 mg/kg zinc sulfate; 273,900 IU/kg vitamin A; 54,780 IU/kg vitamin D; 1,267 IU/kg vitamin E.

milked thrice daily at approximately 0330, 1100, and 1830 h. The sample size was determined to achieve 80% power with an α of 0.05 to detect a 0.15 mM decrease in blood ionized Ca (iCa) concentration in response to EGTA treatment, assuming a standard deviation of 0.01 mM, and accounting for potential animal loss during the transition period (which is typically 25%), resulting in $n = 15$ per treatment.

Thirty multiparous Holstein cows, 21 d before the expected due date (3.17 ± 0.2 parity for the following lactation), were blocked by parity, previous lactation's milk production, and gestation day in a randomized complete block design (Figure 1). They were randomly assigned using a random number generator to receive a continuous 6-h intravenous infusion of either (1) 0.9% sterile NaCl (CON; $n = 15$) or (2) 5% EGTA solution dissolved in 0.9% NaCl (EGTA; $n = 15$), aiming to maintain iCa between 0.7 to 1.0 mM. The 5% EGTA intravenous solution (#E3889, Sigma Aldrich, St. Louis, MO) was prepared according to protocols outlined by Amundson et al. (2018) and Connelly et al. (2022).

Eight days before expected parturition, bilateral jugular vein catheterization was performed. The procedure took place in a veterinary room equipped with a cow chute, where the cows were appropriately restrained. Sedation was performed using 0.02 mg/kg of xylazine (Xylamed, 100 mg/mL, VetOne, Boise, ID). Before catheter insertion, the jugular vein area was shaved using clippers

Table 2. Analyzed nutrient composition (% DM unless otherwise noted) for dry and lactating diets

Nutrient ¹	Diet	
	Dry	Lactating
DM (%)	43.87	42.06
pH	5.01	4.76
CP	14.80	17.37
AD-ICP (% CP)	9.07	5.28
Protein Sol. (% CP)	37.14	54.54
ADF	27.77	20.63
aNDF	40.25	25.03
aNDFom	39.26	24.02
Lignin (% NDFom)	4.23	3.10
uNDFom240	13.07	7.70
NDFD240 (% NDFom)	66.69	67.93
WSC (sugar)	6.99	3.09
ESC (sugar)	4.15	2.38
Starch	22.38	26.48
EE (fat)	3.92	5.57
TFA (fat)	3.02	3.42
Ash	7.15	8.40
Ca	0.74	1.04
P	0.39	0.38
Mg	0.39	0.42
K	1.23	1.57
S	0.36	0.25
Na	0.13	0.42
Cl	0.30	0.54
Adj_CP	14.80	17.37
NFC	36.61	47.63
TDN_OARDC	67.42	75.63
NEL3x_OARDC (Mcal/cwt)	69.63	78.77
ME_OARDC (Mcal/cwt)	112.91	124.04
NEG_OARDC (Mcal/cwt)	45.10	53.57
NEM_OARDC (Mcal/cwt)	72.42	82.04
DCAD (mEq/100 g)	6.20	27.99
RDM (%)	95.11	93.93
ME_higgs (Mcal/cwt)	6.87	9.37

¹AD-ICP = acid detergent insoluble protein; Protein Sol. = soluble protein; aNDF = acid neutral detergent fiber; aNDFom = amylase-treated neutral detergent fiber on an organic matter basis; NDFom = neutral detergent fiber on an organic matter basis; uNDFom = undigested neutral detergent fiber on an organic matter basis; NDFD240 = undigested neutral detergent fiber at 240 hours; WSC = water soluble carbohydrates; ESC = ethanol soluble carbohydrates; EE = ether extract; TFA = *trans* fatty acids; Adj CP = adjusted crude protein; TDN = total digestible nutrients; OARDC = Ohio Agricultural Research and Development Center; cwt = hundred weight; RDM = Redmond real salt/clay; ME_higgs (metabolizable energy) in the Cornell Net Carbohydrate and Protein System.

(PowerPro Ultra Clipper, Oster Pro, Boca Raton, FL) and prepared by scrubbing with betadine (Avrio Health L.P., Stamford, CT) and 70% ethanol 3 times. Local anesthesia was administered via subcutaneous injection of 5.0 mL of 2% lidocaine hydrochloride (Lidocaine Injectable, Aspen Veterinary Resources Ltd., Loveland, CO), followed by an additional cleansing with betadine and 70% ethanol. A commercial kit (Antimicrobial Catheter with Guidewire in Housing, Mila International Inc., Florence, KY) containing a catheter (16 gauge and 200 mm length) equipped with a guidewire (0.81 mm in diameter and 600 mm in length) was used. A small incision (approximately

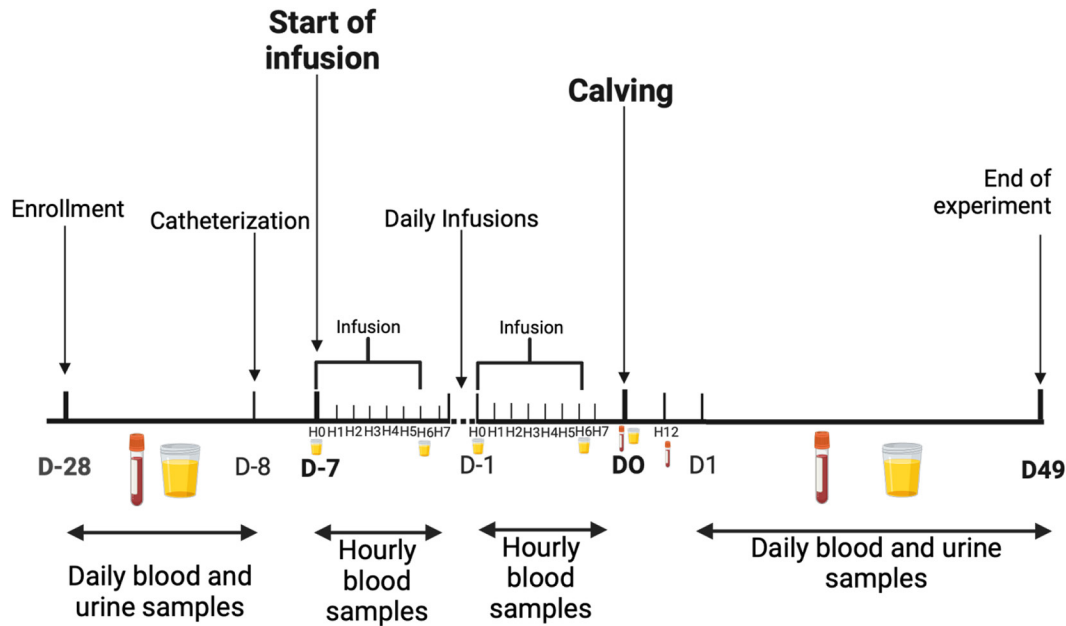


Figure 1. Schematic representation of experimental design and sampling timeline. Thirty dry multiparous Holstein cows were blocked by parity, milk production, and gestation day in a randomized complete block design and assigned to 6 h of intravenous infusions of (1) 5% EGTA ($n = 15/\text{treatment}$) or (2) saline ($n = 15/\text{treatment}$) for at least 3 d and a maximum of 14 d, until parturition.

1 cm) was made using a scalpel no. 10, and the jugular vein was punctured using a 14-gauge and 38-mm-length needle. Subsequently, the guidewire was threaded through the needle into the vein, and upon needle removal, the catheter was passed over the guidewire and positioned within the vein. Following guidewire removal, the catheter was capped and flushed with heparinized saline solution (10,000 IU per 1 L of 0.9% sodium chloride, Hospira Inc., Lake Forest, IL). Three simple sutures were used to secure the catheter to the skin, and the incision site was treated with triple antibiotic ointment (Cosette Pharmaceuticals, NDC 0713–0268–31). Finally, the neck was wrapped with an Elastikon bandage for stabilization. Catheters were removed 12 h postpartum.

Infusions began 7 d before the expected day of parturition and persisted until parturition, spanning 6 h. Cows received a minimum of 3 d of infusion and were removed from the experiment after a maximum of 14 d. If a cow did not fit the criteria, a new cow was included in the study to ensure adequate cow numbers. A total of 13 cows were excluded from the study (7 CON and 6 EGTA). In the CON group, 2 cows were excluded for receiving more than 14 d of infusion, 3 for receiving less than 3 d of infusion, 1 due to jugular catheter loss, and 1 due to death. In the EGTA group, 3 cows were excluded for receiving more than 14 d of infusion and 3 for receiving less than 3 d of infusion.

An infusion pump (Heska Vet IV) was used to maintain and control the IR for the EGTA group. The IR for

EGTA cows was determined based on iCa concentration, and hourly adjustments were made accordingly. Due to different parturition dates, we were unable to pair EGTA and CON. Therefore, CON cows received a constant IR of 500 mL/h. Cows induced into SCH on the first and second infusion days had IR beginning at a rate of 500 mL/h until blood iCa fell below 1.0 mM. From the third infusion day onward, IR were initiated at 1,000 mL/h due to challenges in inducing SCH. Blood iCa concentration was measured using a handheld biochemical analyzer and analyzed using the CG8+ cartridge (i-Stat System, Abbott Laboratories).

Blood samples were collected daily from the coccygeal vein or artery from enrollment to the onset of infusion and from 1 to 49 DIM using a 20-gauge Vacutainer needle (Greiner Bio-One GmbH, Exelint International Co.). Throughout infusions, blood samples were collected from the jugular catheter immediately before infusion started, hourly during infusion (from the contralateral site of infusion), and 1 h post-infusion termination, using a 20-mL syringe and promptly transferring to evacuated tubes. Before blood sampling from the catheter, 10 mL of blood was discarded, and after the collection, the catheter was washed with 10 mL of heparin solution (10,000 IU per 1 L of 0.9% sodium chloride, Hospira Inc., Lake Forest, IL). Blood samples were collected immediately after calving and 12 h later.

Whole blood was collected into 10-mL BD Vacutainer K2 EDTA Plus (368589, Becton, Dickinson, and Com-

pany), 10-mL BD Vacutainer Serum (367820, Becton, Dickinson, and Company), and 10-mL Lithium Heparin 158 USP Units (367880, Becton, Dickinson, and Company) blood collection tubes and gently inverted. Immediately after inversion, approximately 1.5 mL of whole blood was transferred from the 10-mL Vacutainer K2 EDTA Plus tube to a 2-mL Eppendorf preloaded with approximately 14 mg of ascorbic acid (10 mg/mL) to prevent serotonin oxidation, and subsequently frozen at -20°C until further analysis (Connelly et al., 2021). Samples from whole blood collected into lithium heparin tubes during the infusion period until 3 DIM were evaluated for blood parameters using a handheld biochemical analyzer and analyzed using the CG8+ cartridge (VetScan i-Stat, Abaxis), with measurements and calculations not temperature-corrected. The parameters analyzed were pH; concentrations of HCO_3^- ; base excess (BE); partial pressure of CO_2 (pCO_2); total dissolved CO_2 (tCO_2); partial pressure of O_2 (pO_2); O_2 saturation; concentrations of sodium (Na), potassium (K), and ionized Ca (iCa); hemoglobin (Hb); and hematocrit. Blood samples collected into serum tubes were left at room temperature for 1 h to clot before centrifugation. Serum and plasma were harvested after centrifugation at $3,000 \times g$ for 20 min at 4°C . Samples were allocated into duplicate aliquots and stored at -80°C until further analysis.

Milk weights were recorded at each milking. Two milk samples were collected weekly (every Monday) during the initial 3 wk of lactation during each milking. One of the milk samples was refrigerated at 4°C and subsequently analyzed at AgSource Laboratories using MilkScan FT+ and MilkoScan 7RM FTIR instruments. The second milk aliquot was frozen at -20°C until further analysis. Body weight measurements were performed once a week. The TMR samples were collected daily, and once a week, the daily samples were composited and a subsample was sent to Dairyland Laboratories Inc. (Arcadia, WI) for near infrared analysis. Data for the calculated 305-d milk production were collected from Bovisync. However, it should be acknowledged that estimates produced during early lactation are known to have limited reliability.

Laboratory Analysis

Serum and milk tCa concentrations were determined using an *o*-cresolphthalein-calcium reaction (700550, Cayman Chemical) per the manufacturer's instructions as previously described (Laporta et al., 2013). Before measurement of milk tCa, samples were mixed with 0.1 M acetic acid in a 1:1 proportion and centrifuged at $13,000 \times g$ for 12 min at 4°C to allow protein precipitation, then the supernatant was diluted in a 1:40 proportion and analyzed for tCa concentration (Connelly et al., 2021; Frizzarini et al., 2024). Serum tCa intra- and interassay

CV were 2.44% and 9.86%, respectively. Milk tCa intra- and interassay CV were 2.84% and 19.20%, respectively.

Serum PTH concentrations were measured on the first infusion day, d -2 , and d -1 , both before the start of the infusion (h 0) and immediately after the end of the infusion (h 6). Samples of PTH were measured in single samples (except for the standard curve and pool, which were analyzed in duplicate on every plate), using a bovine Intact PTH ELISA kit (DEIA 1826B, Creative Diagnostics) per the manufacturer's instructions. The PTH interassay CV was 4.45%. 5-HT whole blood concentrations were measured at the same sample times as PTH. Blood 5-HT was measured in singles (except for the standard curve and pool, which were analyzed in duplicate on every plate), due to the high reliability and reproducibility of the assay. The procedure followed the manufacturer's instructions (IM1749, Immunotech, Beckman Coulter) as previously described (Connelly et al., 2020), and the interassay CV was 7.48%.

Statistical Analysis

Prepartum and postpartum samples were analyzed separately. The prepartum data were restricted from the first infusion day, and d -3 to d -1 (d 3, 2, and 1 before partum) for analysis. All statistical analyses were performed using the MIXED procedure of SAS (version 9.4, SAS Institute Inc., Cary, NC). A baseline measurement was determined using the first sample collected for each cow upon their transfer to the tiestall barn, serving as a covariate for respective analysis.

Prepartum cows at 254 d of gestation were blocked by lactation number (lactation 2 vs. lactation >3) and expected due date. For serum tCa, blood gases, minerals, PTH, and 5-HT, we fitted linear mixed-effects models. For the prepartum period, the model included fixed effects of treatment, lactation, block, day (expected day prepartum or day postpartum), infusion time (time), and their interactions (treatment $\times$ day, treatment $\times$ time, and treatment $\times$ day $\times$ time), as well as the covariate. For the postpartum period, the same variables (tCa, blood gases, minerals, PTH, and 5-HT) were analyzed with fixed effects including treatment, lactation, block, day, treatment $\times$ day, and covariate.

For cows in the EGTA treatment group, the IR data (mL of EGTA solution per minute required to sustain subclinical hypocalcemia) were analyzed to assess variation across infusion days and time points, with fixed effects including lactation, block, day, time, and day $\times$ time interaction. Milk components and ECM data were restricted from wk 1 to 4, whereas tCa milk data were limited from wk 1 to 4. These analyses included fixed effects for treatment, week, lactation, block, treatment $\times$ week, and covariate. Additionally, fixed effects for milk

production included treatment, lactation, block, day, treatment $\times$ day, and covariate, whereas total 305-d milk production was analyzed with fixed effects for treatment, lactation, and block.

In all models, cow and time (or day or week) were included as repeated measures. An autoregressive structure of order 1 [ar(1)] was used for prepartum tCa, prepartum blood gases and minerals, IR, tCa in milk, milk components, PTH, and 5-HT. For postpartum tCa, postpartum blood gases and minerals, milk production, PTH, and 5-HT, a spatial power covariance structure was applied due to different sampling timeframes.

Model residuals were tested for normality, and when assumptions were not met, data were transformed according to diagnostic plots. Results are presented as LSM $\pm$ SEM for normally distributed variables, or as mean (range) for data that did not meet normality. Statistical significance was declared if $P \leq 0.05$. Mean comparisons were performed using the Tukey test when significant “treatment” or “treatment $\times$ time” or “treatment $\times$ day” or “treatment $\times$ day $\times$ time” effects were detected.

RESULTS

Blood Gases, Minerals, Acid-Base Status, and Infusion Rate

Blood gas parameters, blood mineral concentrations, and blood acid-base status during prepartum and postpartum periods for cows treated with saline or 5% EGTA are presented in Table 3. A significant treatment effect was observed for blood pH during the prepartum period ($P < 0.01$), with cows in the EGTA group exhibiting lower values compared with the CON group (7.39 ± 0.0 and 7.42 ± 0.0 , respectively; Table 3). Conversely, during the postpartum period, cows in the EGTA group had higher blood pH compared with the CON group ($P = 0.05$; 7.44 ± 0.0 and 7.42 ± 0.0 , respectively; Table 3). A treatment effect was also observed for pCO₂ during the prepartum period ($P = 0.02$), with cows in the EGTA group exhibiting lower concentrations compared with the CON group (38.84 ± 0.6 and 40.83 ± 0.6 mm Hg, respectively; Table 3). A similar treatment effect was observed for tCO₂ during the prepartum period ($P < 0.01$), with cows in the EGTA group having reduced concentrations compared with the CON group (24.67 ± 0.4 and 27.60 ± 0.5 mM, respectively; Table 3). Significant treatment effects were also observed for HCO₃⁻ ($P < 0.01$) and BE⁻ ($P < 0.01$). Cows in the EGTA group had reduced HCO₃⁻ concentrations (23.54 ± 0.4 vs. 26.35 ± 0.5 mM) and reduced BE (-1.54 ± 0.5 vs. 1.88 ± 0.5 mM) compared with the CON group (Table 3). Additionally, cows in the EGTA group had elevated hemoglobin concentrations (9.45 ± 0.1 vs. 9.05 ± 0.1 g/dL; $P < 0.01$) compared with the CON group (Table 3).

For iCa, there was a treatment effect during both the prepartum and postpartum periods ($P < 0.01$ for both periods). During the prepartum period, a 3-way interaction between treatment $\times$ day $\times$ time for iCa concentrations was observed, with cows in the EGTA group exhibiting reduced iCa concentrations from h 1 to 7 on the first day of infusion, and d -3 to -1 ($P = 0.02$; Table 3 and Figure 2B). During the postpartum period, there was an interaction between treatment $\times$ day ($P < 0.01$) for iCa concentrations, with cows in the EGTA group exhibiting higher iCa concentrations on d 0 h 12 (12 h after calving), d 1, and d 2 (Figure 2D).

Blood tCa concentration during the prepartum period according to treatment, day, and time is presented in Figure 2A. Significant treatment ($P < 0.01$), day ($P < 0.01$), and time ($P < 0.01$) effects were observed on blood tCa concentrations during the prepartum period. As expected, a significant treatment $\times$ time interaction was observed during the prepartum period ($P < 0.01$), with the EGTA group exhibiting reduced blood tCa concentrations from h 1 to 7. During the postpartum period, significant day ($P < 0.01$) and treatment $\times$ day interaction ($P < 0.01$) effects were observed. Cows in the EGTA group had increased blood tCa concentrations on d 0, d 0 h 12, d 1, d 2, d 3, and d 5 compared with the CON group (1.96 ± 0.1 vs. 1.60 ± 0.1 , 2.16 ± 0.1 vs. 1.52 ± 0.1 , 2.16 ± 0.1 vs. 1.64 ± 0.1 , 2.18 ± 0.1 vs. 1.81 ± 0.1 , 2.40 ± 0.1 vs. 2.11 ± 0.1 , and 2.53 ± 0.1 vs. 2.21 ± 0.1 mmol/L, respectively; Figure 2C).

Infusion rates for the EGTA group during the prepartum period are presented in Figure 2E. Significant effects for day ($P < 0.01$), time ($P < 0.01$), block ($P < 0.01$), and day $\times$ time interaction ($P < 0.01$) were observed. The highest IR was observed on d -3 (858.48 ± 39.85 mL/h; Figure 2E) and the lowest on the first day of infusion (622.68 ± 40.19 mL/h; Figure 2E). The IR was the lowest on the first infusion day at h 0 and h 1, and on d -1 at h 4 (494.71 ± 62.7 , 483.99 ± 62.7 , and 527.41 ± 62.5 mL/h, respectively). The IR was the highest on d -1 at h 0, and on d -2 at h 0 and h 1 ($1,005.98 \pm 62.5$, $1,005.98 \pm 62.5$, and 970.27 ± 62.5 mL/h, respectively).

Serum PTH concentrations during the prepartum period according to treatment, day, and time are presented in Figure 3A. There was an overall treatment effect ($P = 0.05$), with EGTA-treated cows exhibiting an increased PTH concentration compared with the CON-treated cows (194.11 ± 10.53 vs. 164.18 ± 11.03 pg/mL, respectively). An effect of day was observed ($P < 0.01$), with the highest PTH concentrations on d -2 and d -1, compared with d -7 (190.32 ± 9.78 , 188.52 ± 9.86 , and 158.60 ± 9.57 pg/mL, respectively). Additionally, we observed an effect of time ($P < 0.01$) with increased PTH concentrations on h 0 compared with h 6 (200.72 ± 8.29 vs. 157.57 ± 8.29 pg/mL, respectively). There was a treatment $\times$

Table 3. Effect of induction of subclinical hypocalcemia during the prepartum period on blood parameters and acid-base status in multiparous Holstein cows during prepartum and postpartum periods (LSM $\pm$ SE)

Item	Treatment ¹			P-value ²							
	CON	EGTA	Trt	Day	Time	Lact	Block	Day × Trt	Time × Trt	Time × Day	Time × Day × Trt
Prepartum ³											
Blood pH	7.42 ± 0.00 ^a	7.39 ± 0.00 ^b	<0.01	0.06	<0.01	0.73	0.33	<0.01	<0.01	0.13	0.15
pCO ₂ ⁴ (mm Hg)	40.83 ± 0.62 ^a	38.84 ± 0.60 ^b	0.02	0.32	<0.01	0.84	0.01	0.24	<0.01	0.34	0.19
Total CO ₂ (mM)	27.60 ± 0.47 ^a	24.67 ± 0.45 ^b	<0.01	0.12	<0.01	0.92	0.57	<0.01	<0.01	0.13	0.81
pO ₂ ⁵ (mm Hg)	38.77 ± 0.96 ^a	41.43 ± 0.90 ^a	0.04	0.02	0.01	0.42	0.02	0.38	0.04	0.61	0.86
HCO ₃ ⁻ (mM)	26.35 ± 0.45 ^a	23.54 ± 0.43 ^b	<0.01	0.07	<0.01	0.87	0.53	<0.01	<0.01	0.10	0.75
Base excess (mM)	1.88 ± 0.51 ^a	-1.54 ± 0.48 ^b	<0.01	0.06	<0.01	0.94	0.04	<0.01	<0.01	0.07	0.84
O ₂ saturation (%)	70.20 ± 1.14 ^b	73.85 ± 1.08 ^a	0.02	0.01	0.01	0.32	0.01	0.55	0.32	0.88	0.88
Glucose (mg/dL)	70.68 ± 1.19	73.74 ± 1.12	0.06	<0.01	<0.01	0.97	0.49	0.97	<0.01	<0.01	0.46
Na (mM)	144.43 ± 0.23 ^b	145.70 ± 0.22 ^a	<0.01	<0.01	<0.01	0.92	0.12	0.44	<0.01	0.39	0.07
K (mM)	3.78 ± 0.04	3.72 ± 0.03	0.20	0.01	<0.01	0.05	0.42	0.50	0.04	0.91	0.26
iCa ⁶ (mM)	1.24 ± 0.02 ^a	0.93 ± 0.01 ^b	<0.01	0.01	<0.01	0.80	0.99	0.03	<0.01	0.02	0.02
Hematocrit (%)	26.62 ± 0.27 ^b	27.79 ± 0.25 ^a	<0.01	<0.01	<0.01	0.45	0.80	0.73	<0.01	0.01	0.11
Hemoglobin (g/dL)	9.05 ± 0.09 ^b	9.45 ± 0.09 ^a	<0.01	<0.01	<0.01	0.52	0.78	0.75	<0.01	0.01	0.1
Postpartum ⁷											
Blood pH	7.42 ± 0.01 ^b	7.44 ± 0.01 ^a	0.05	<0.01	—	0.35	0.82	0.57	—	—	—
pCO ₂ ⁴ (mm Hg)	47.33 ± 0.96	47.01 ± 0.92	0.81	0.33	—	0.85	0.75	0.76	—	—	—
Total CO ₂ (mM)	31.89 ± 0.58	33.00 ± 0.55	0.18	<0.01	—	0.44	0.94	0.28	—	—	—
pO ₂ ⁵ (mm Hg)	41.03 ± 3.92 ^b	51.79 ± 3.64 ^a	0.04	0.83	—	0.54	0.85	0.78	—	—	—
HCO ₃ ⁻ (mM)	30.35 ± 0.53	31.53 ± 0.50	0.11	<0.01	—	0.43	0.79	0.54	—	—	—
Base excess (mM)	5.81 ± 0.59	7.36 ± 0.55	0.07	<0.01	—	0.34	0.97	0.21	—	—	—
O ₂ saturation (%)	64.53 ± 2.82	72.23 ± 2.62	0.06	0.99	—	0.52	0.96	0.79	—	—	—
Glucose (mg/dL)	70.32 ± 2.53	71.55 ± 2.28	0.73	<0.01	—	0.29	0.53	0.61	—	—	—
Na (mM)	143.73 ± 0.29 ^a	142.33 ± 0.27 ^b	<0.01	<0.01	—	0.93	0.47	<0.01	—	—	—
K (mM)	4.21 ± 0.08	4.20 ± 0.08	0.88	<0.01	—	0.61	1.00	0.74	—	—	—
iCa ⁶ (mM)	0.99 ± 0.03 ^b	1.12 ± 0.02 ^a	<0.01	<0.01	—	0.01	0.16	<0.01	—	—	—
Hematocrit (%)	27.73 ± 0.44	27.15 ± 0.42	0.37	<0.01	—	0.58	0.05	0.27	—	—	—
Hemoglobin (g/dL)	9.43 ± 0.15	9.23 ± 0.14	0.37	<0.01	—	0.57	0.05	0.31	—	—	—

^{a,b}Values with different superscripts within a row differ.¹Prepartum cows were treated with saline (CON) or 5% EGTA (EGTA) infusions for 6 h from d -7 (relative to the expected due date) until d -1.²Trt = treatment; Time = time relative to prepartum infusion; Lact = lactation.³Prepartum samples were collected on the first infusion day and d -3 to -1 relative to calving.⁴pCO₂ = partial pressure of CO₂.⁵pO₂ = partial pressure of O₂.⁶iCa = ionized calcium.⁷Postpartum samples were collected immediately after calving, 12 h after calving, and d 1 to 3.

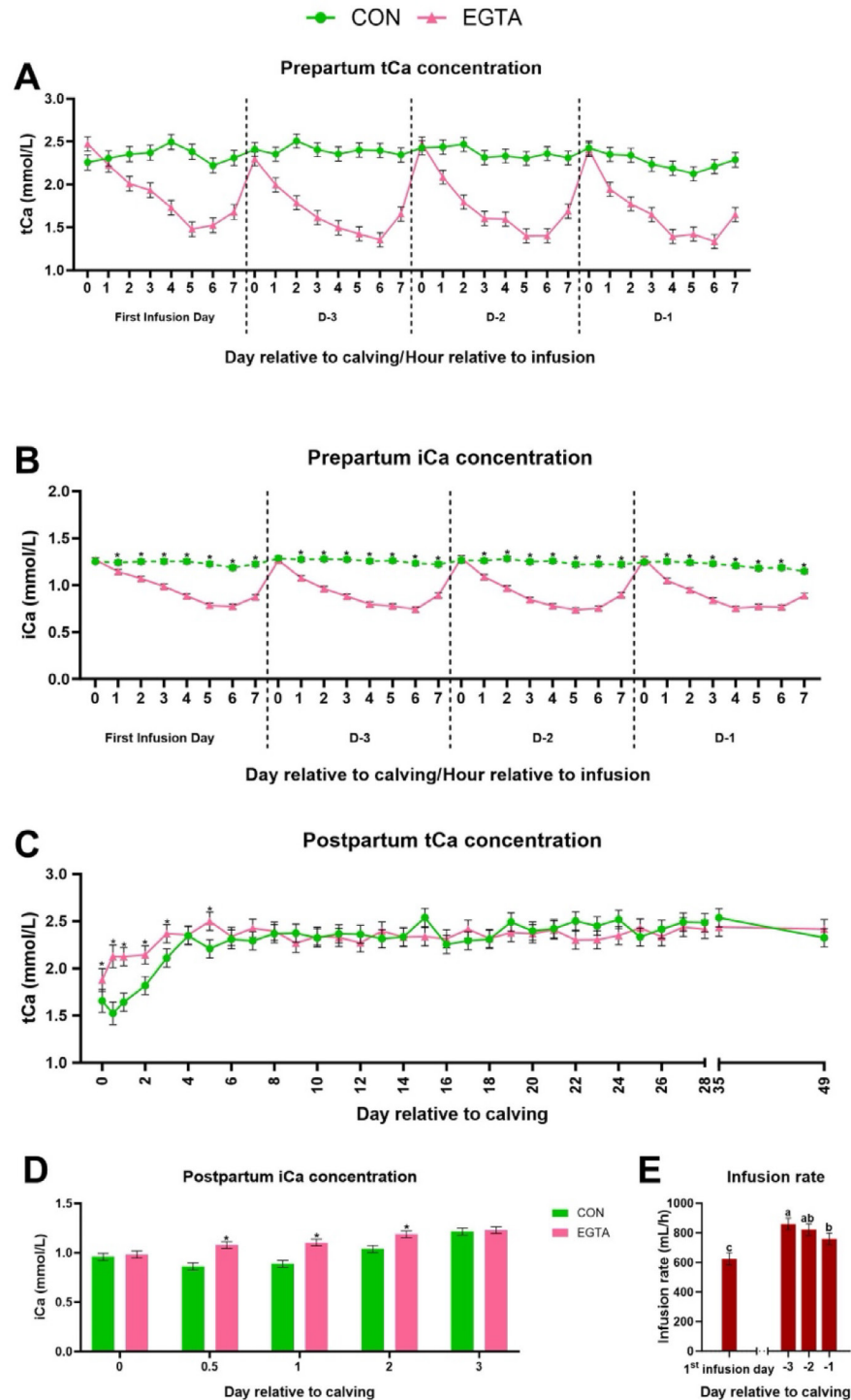


Figure 2. Least squares means $\pm$ SE for (A) prepartum tCa (mmol/L), (B) prepartum iCa (mmol/L), (C) postpartum tCa (mmol/L), (D) postpartum iCa (mmol/L), and (E) infusion rate (mL/h) for multiparous cows infused with saline (CON) or 5% EGTA (EGTA) for a minimum of 4 d and a maximum of 14 d prepartum. Prepartum and postpartum data were analyzed separately. There was a treatment effect ($P < 0.01$), day effect ($P < 0.01$), time effect ($P < 0.01$), and treatment by time effect ($P < 0.01$) during the prepartum for tCa. There was a treatment effect ($P < 0.01$), day effect ($P < 0.01$), time effect ($P < 0.01$), day $\times$ treatment effect ($P = 0.03$), time $\times$ treatment effect ($P < 0.01$), time by day effect ($P = 0.02$), and treatment by day by time effect ($P = 0.02$) for prepartum iCa. There was a day effect ($P < 0.01$), and treatment by day effect ($P < 0.01$) for postpartum tCa. There was a treatment effect ($P < 0.01$), a day effect ($P < 0.01$), a lactation effect ($P = 0.01$), and a treatment $\times$ day effect ($P < 0.01$) for postpartum iCa. There was a day effect ($P < 0.01$), a time effect ($P < 0.01$), a block effect ($P < 0.01$), and day $\times$ time effect ($P < 0.01$). *Treatment $\times$ day $\times$ time effect or treatment $\times$ day effect when treatment means differed significantly ($P \leq 0.05$). (a–c) Different lowercase letters depict treatment $\times$ day $\times$ time effects when day means differed significantly ($P \leq 0.05$).

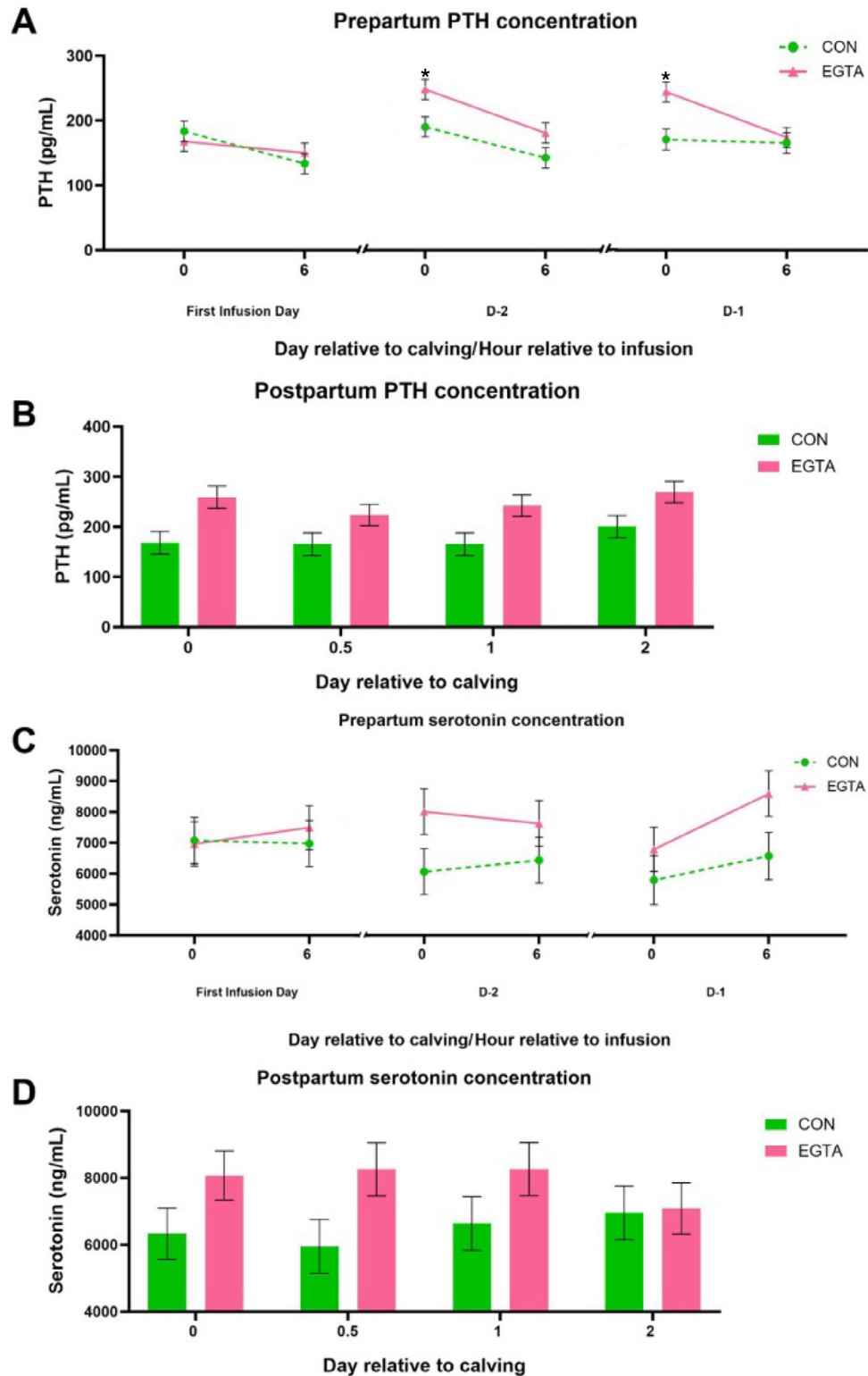


Figure 3. Least squares means $\pm$ SE for (A) prepartum PTH (pg/mL), (B) postpartum PTH (pg/mL), (C) prepartum 5-HT (ng/mL), and (D) postpartum 5-HT (ng/mL) for multiparous cows infused with saline (CON) or 5% EGTA (EGTA) for a minimum of 4 d and a maximum of 14 d prepartum. Prepartum and postpartum data were analyzed separately. There was a treatment effect ($P = 0.05$), day effect ($P < 0.01$), time effect ($P < 0.01$), treatment $\times$ day effect ($P = 0.05$), and treatment $\times$ day $\times$ time effect ($P = 0.01$) during the prepartum period for PTH. There was a treatment effect ($P = 0.01$) and a day effect ($P = 0.04$) for postpartum PTH. There was a treatment tendency ($P = 0.06$) for prepartum 5-HT. There was a treatment $\times$ day tendency ($P = 0.09$) for postpartum 5-HT. *Treatment $\times$ day $\times$ time effect when treatment means differed significantly ($P \leq 0.05$).

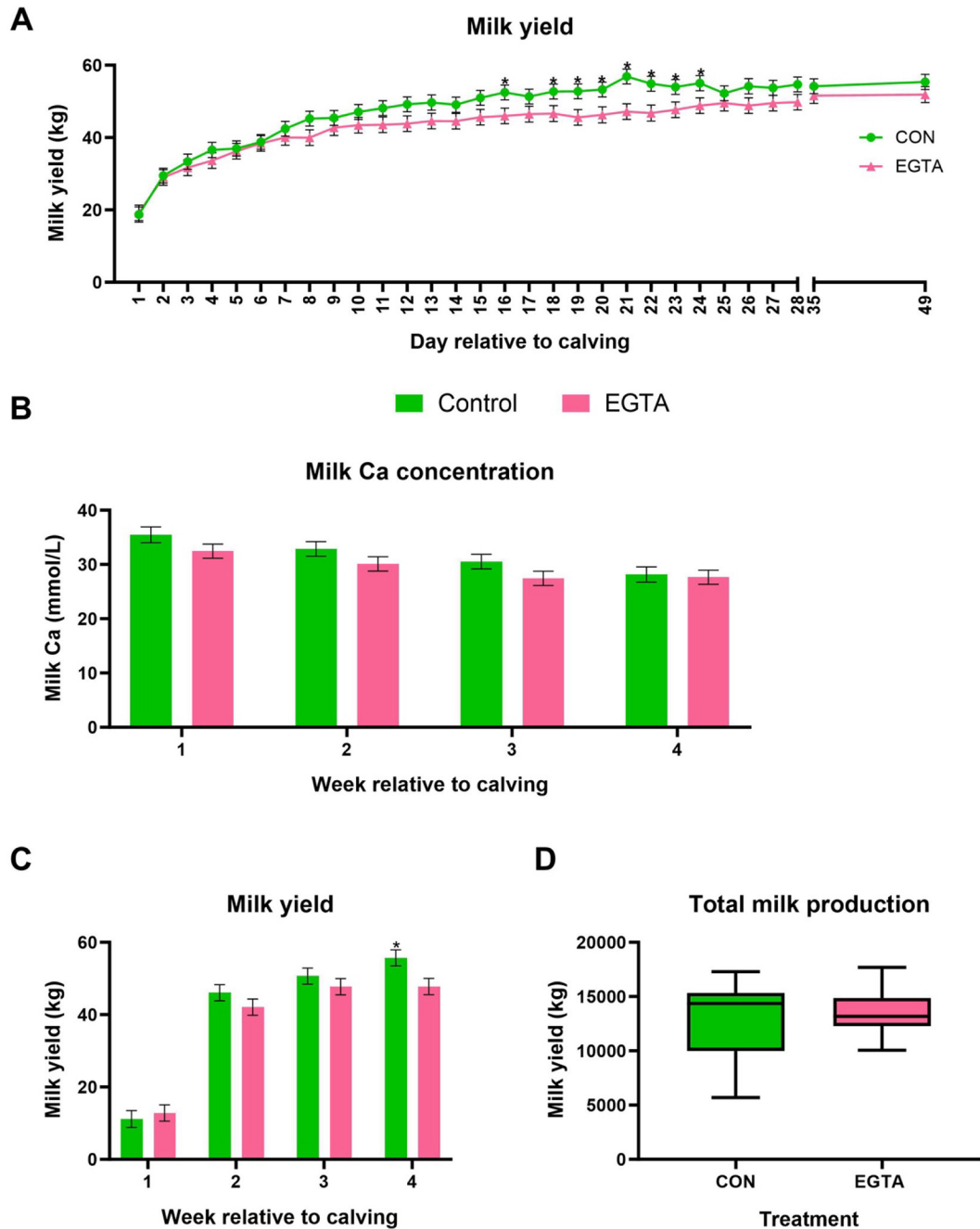


Figure 4. Least squares means $\pm$ SE for (A) daily milk production (kg), (B) milk Ca concentration (mmol/L), (C) weekly milk production (kg), and (D) total milk production (kg) for multiparous cows infused with saline (CON) or 5% EGTA (EGTA) for a minimum of 4 d and a maximum of 14 d prepartum. Lower and upper whiskers reflect the minimum and maximum values, the midline is the median, and the box represents the 25th and 75th quartiles. There was a day effect ($P < 0.01$) and treatment $\times$ day effect ($P < 0.01$) for daily milk production. There was a week effect ($P < 0.01$) for milk Ca concentration. There was a week effect ($P < 0.01$) and treatment $\times$ week effect ($P = 0.02$) for weekly milk production. *Treatment $\times$ day effect or treatment $\times$ week effect when treatment means differed significantly ($P \leq 0.05$).

day $\times$ time interaction ($P = 0.01$), with the EGTA group exhibiting increased PTH concentrations on d -2 h 0 (246.24 ± 15.23 vs. 188.74 ± 15.87 pg/mL) and on d -1 h 0 (242.45 ± 15.23 vs. 168.66 ± 16.28 pg/mL). There was a treatment $\times$ day interaction ($P = 0.05$), with EGTA-

treated cows having the highest PTH concentrations on d -2 and d -1 (214.41 ± 13.67 and 209.05 ± 13.67 pg/mL, respectively) compared with control cows. Serum PTH concentration during the postpartum period according to treatment and day is presented in Figure 3B. There was

Table 4. Effect of induction of subclinical hypocalcemia during the prepartum period on milk production and composition and DMI in multiparous Holstein cows (LSM $\pm$ SE, or mean [range])

Item	Treatment ¹		P-value ²						
	CON	EGTA	Trt	Week	Lact	Block	Trt × Week	Day	Trt × Day
Milk production (kg)	40.90 ± 1.83	37.60 ± 1.84	0.18	<0.01	0.76	0.12	0.02	—	—
Fat (%)	4.85 ± 0.17	4.79 ± 0.17	0.81	<0.01	0.39	0.74	0.34	—	—
Fat (g)	191.96 ± 10.08	173.91 ± 10.07	0.18	<0.01	0.82	0.08	0.01	—	—
Protein (%)	3.36 ± 0.05	3.25 ± 0.05	0.15	<0.01	0.16	0.43	0.37	—	—
Protein (g)	131.04 ± 5.60	117.94 ± 5.60	0.08	<0.01	0.82	0.02	0.05	—	—
Lactose (%)	4.69 ± 0.03	4.71 ± 0.03	0.62	<0.01	0.31	0.88	0.39	—	—
Lactose (g)	194.27 ± 8.74	179.57 ± 8.76	0.21	<0.01	0.87	0.17	0.05	—	—
SNF (%)	9.28 ± 0.07	9.15 ± 0.07	0.16	<0.01	0.11	0.15	0.07	—	—
SNF (g)	373.20 ± 16.15	340.52 ± 16.18	0.13	<0.01	0.98	0.07	0.05	—	—
SCC (1,000 cells/mL)	100.96 [10.33, 5,028.33]	128.14 [13.33, 2,230.00]	0.15	<0.01	0.34	0.51	0.84	—	—
MUN (mg/dL)	12.03 ± 0.44	11.01 ± 0.44	0.09	0.72	0.53	<0.01	0.81	—	—
305-d milk production (kg)	13,677 ± 512.49	13,242 ± 482.70	0.54	—	0.10	0.67	—	—	—
ECM (kg)	45.70 ± 1.98	42.24 ± 1.91	0.18	<0.01	0.42	0.38	0.02	—	—
Prepartum DMI (kg)	13.32 ± 0.53	12.91 ± 0.53	0.56	—	0.11	0.85	—	<0.01	0.14
Postpartum DMI (kg)	21.31 ± 0.99	21.32 ± 1.01	0.99	—	0.23	0.69	—	<0.01	0.40

¹Prepartum cows were treated with saline (CON) or 5% EGTA (EGTA) infusions for 6 h from d -7 (relative to the expected due date) until d -1.²Trt = treatment; Week = week relative to calving; Lact = lactation.

a treatment effect ($P = 0.01$), with EGTA-treated cows exhibiting overall increased PTH concentrations compared with CON-treated cows (248.65 ± 17.21 vs. 174.93 ± 18.29 pg/mL, respectively). There was a day effect ($P = 0.04$), with the highest PTH concentrations occurring on d 2 (234.90 ± 15.56).

Whole blood 5-HT concentration during the prepartum period according to treatment, day, and time is presented in Figure 3C. The EGTA-treated cows had an overall numerically higher 5-HT concentration compared with the CON-treated cows ($7,761.60 \pm 461.04$ vs. $6,409.59 \pm 482.80$ ng/mL, respectively; $P = 0.06$). Whole blood 5-HT concentration during the postpartum period according to treatment and day is presented in Figure 3D. The EGTA-treated cows on d 1 had numerically the highest 5-HT concentration, and CON-treated cows on d 0 h 12 had numerically the lowest concentration ($8,264.47 \pm 794.58$ vs. $5,950.31 \pm 804.33$ ng/mL, respectively; $P = 0.09$).

Production Variables

Daily milk production by treatment and day is presented in Figure 4A. Significant effects were observed for day ($P < 0.01$) and the treatment $\times$ day interaction ($P < 0.01$). The highest milk production was recorded on d 49 (53.62 ± 1.6 kg), and the lowest on d 0 h 12 (9.02 ± 1.6 kg). Cows in the EGTA group exhibited lower milk production on d 16, and d 18 to d 24 compared with CON cows (Figure 4A). Total milk Ca concentration according to treatment and week is presented in Figure 4B. A significant week effect was observed ($P < 0.01$), with the highest milk tCa concentration observed on wk 1 (33.95 ± 0.98 mM), and the lowest on wk 3 and 4 (28.98 ± 0.95 and 27.90 ± 0.95 , respectively).

Weekly milk production (Figure 4C) and composition are summarized in Table 4. A significant treatment $\times$ week interaction effect was observed for milk production ($P = 0.02$), fat (g; $P = 0.01$), protein (g; $P = 0.05$), lactose (g; $P = 0.05$), and SNF ($P = 0.05$). Cows in the EGTA group produced less milk in wk 4 compared with cows in the CON group (47.77 ± 2.2 vs. 55.70 ± 2.2 kg). Additionally, cows in the EGTA group produced milk with reduced fat yield in wk 2 and 4 compared with cows in the CON group (191.36 ± 13.2 vs. 235.45 ± 13.1 , 199.88 ± 13.2 vs. 235.95 ± 13.1 g, respectively). Protein yield was also decreased in milk from EGTA-treated cows on wk 4 compared with CON cows (138.39 ± 7.2 vs. 163.42 ± 7.1 g), as was lactose yield (231.42 ± 10.7 vs. 266.95 ± 10.7 g) and SNF (420.38 ± 20.2 vs. 490.08 ± 20.1 g). There was no treatment effect for total milk production for 305 d ($13,677 \pm 512.5$ and $13,242 \pm 482.7$ kg for CON and EGTA, respectively; Table 4 and Figure 4D). There was also no treatment or treatment $\times$ week interaction effects for DMI during the prepartum (13.32 ± 0.5

and 12.91 ± 0.5 kg for CON and EGTA, respectively) and postpartum period (21.31 ± 1.0 and 21.32 ± 1.0 kg for CON and EGTA, respectively; Table 4).

DISCUSSION

Dairy cows experience a substantial demand for Ca during early lactation, as the mineral content in milk and colostrum can be 10 and 20 times higher, respectively, than the Ca concentration in blood (Horst et al., 2005). Despite various dietary strategies and Ca supplementation postpartum, SCH remains prevalent. Recent literature suggests that SCH is essential for stimulating the Ca feedback loop to maintain Ca homeostasis (McArt and Neves, 2020; Connelly et al., 2024). Our data further support the role of SCH in facilitating the physiological adaptations required for lactation, enhancing the cow's ability to manage Ca perturbations during early lactation.

Cows treated with EGTA during the prepartum period exhibited a consistent Ca pattern across all infusion days. At h 0, there were no differences between treatments; however, Ca concentrations in the EGTA group decreased with the start of the infusion. Notably, by h 7, just 1 h after the infusion termination, Ca concentrations in the EGTA group began to rise, returning to similar concentrations with the CON group by the next day's h 0 measurement. These results showed that EGTA-treated cows had adapted to low Ca concentrations, likely mobilizing Ca from bone, gut, and kidney to maintain Ca homeostasis (Connelly et al., 2022).

Interestingly, IR was the lowest on the first infusion day and increased on d -3 and d -2, suggesting cows required higher EGTA infusion rates to sustain SCH. This aligns with the findings by Connelly et al. (2022), who reported that early-lactation cows needed more EGTA to induce SCH compared with nonlactating cows. Comparing the IR between the present study and Connelly et al. (2022), we observed that prepartum cows had higher IR than early-lactation cows. This result suggests that prepartum cows had a more robust adaptation to support blood Ca homeostasis than early-lactating cows, who required lower IR to maintain SCH.

Webster et al. (2025) found that prepartum EGTA infusion increased iCa and tCa postpartum. Confirming our observations in Webster et al. (2025) and our primary hypothesis, cows receiving prepartum EGTA exhibited higher Ca concentrations from d 0 h 12 to d 2 and again on d 4, indicating better adaptation to the increased Ca demand after parturition. This supports the notion that SCH activates endocrine mechanisms to compensate for the Ca chelation by mobilizing the internal Ca reservoir in the body (Connelly et al., 2022). Further confirmation of this is the results in Connelly et al. (2024), who observed that cows receiving Ca infusions for 24 h starting

immediately after calving had higher blood Ca concentrations than those receiving dextrose (Connelly et al., 2024). However, after the end of the infusion period, Ca concentrations dropped below the dextrose group and remained lower for 24 h after the infusion (Connelly et al., 2024). Collectively, these studies underscore the importance of SCH-induced activation of Ca regulatory mechanisms for maintaining Ca homeostasis.

In a previous study, the authors fed cows with high Ca (87.2 g/d) or low Ca (26.4 g/d) for 6 wk prepartum (van de Braak et al., 1987). After 10 h postparturition, they initiated a Na_2EDTA infusion, which lasted until Ca concentration was about 1.0 mmol/L (van de Braak et al., 1987). The authors found that after the Na_2EDTA infusion, cows receiving the low Ca diet prepartum recovered better than those receiving high Ca diets (van de Braak et al., 1987). Low Ca diet cows had higher PTH concentrations, and the authors speculated that PTH might have stimulated 1-hydroxylase, the enzyme responsible for the synthesis of 1,25-dihydroxycholecalciferol, thereby increasing Ca absorption in the gut. In our study, we found that EGTA-treated cows had higher PTH concentrations, consistent with EGTA's strong affinity for chelating iCa (Connelly et al., 2022). Specifically, cows treated with EGTA prepartum experienced a 42% increase in PTH concentration postpartum, which corresponds with their increased blood Ca concentration, supporting Connelly et al.'s (2022) hypothesis.

Previous studies showed that administration of 5-HTP during lactation induced a transient hypocalcemia along with increased Ca concentration in the MG (Laporta et al., 2015; Connelly et al., 2021). However, when EGTA-induced Ca chelation occurred during prepartum or early lactation, it did not affect 5-HT concentrations, suggesting the serotonin-Ca axis does not act as a feedback loop (Connelly et al., 2022; Webster et al., 2025). In contrast, in our study, using the same EGTA method as Connelly et al. (2022), and more cows than Webster et al. (2025), we observed both decreased blood Ca and increased 5-HT concentrations, indicating a potential regulatory role of serotonin in response to hypocalcemia during the prepartum period.

Similar to previous studies, EGTA or Na_2EDTA infusions induced metabolic acidosis, evidenced by decreased blood pH, tCO_2 , HCO_3^- , and BE (Desmecht et al., 1995; Connelly et al., 2022). The observed increase in Hb concentrations in EGTA-treated cows during the prepartum period is consistent with earlier findings where Na_2EDTA infusions in calves led to systemic Hb increases, likely due to hemolysis caused by Ca-EDTA complex formation (Desmecht et al., 1995). The reduction in pCO_2 aligns with the compensatory respiratory alkalosis response to metabolic acidosis; for every 1 mEq/L drop of HCO_3^- , there is a decrease of 1.2 mEq/L in pCO_2 (Adrogué and Madias, 2010; Hall and Guyton, 2011).

Previous research has also linked decreased Ca concentrations in d 1 with increased milk production (Neves et al., 2018; Seely et al., 2021). The initial transient drop in blood Ca concentrations immediately after parturition is suggested to enhance PTHLH response, aiding cows in meeting the metabolic demands of lactation and preparing them for higher milk production (Neves et al., 2018). McArt and Neves (2020) also conducted a study examining the effects of SCH on milk production. They observed that multiparous cows experiencing transient SCH, characterized by low Ca concentration on d 1 with recovery by d 4, produced more milk compared with those experiencing delayed SCH, marked by high Ca concentrations on d 1 followed by a decline by d 4 (McArt and Neves, 2020). The authors suggested that cows undergoing transient SCH had an improved transition into lactation, as evidenced by their higher milk yield, compared with cows that did not exhibit this initial drop in Ca concentrations (McArt and Neves, 2020). Additionally, they reported an increased incidence of health events and a higher risk of culling among cows experiencing persistent (low Ca concentrations on both d 1 and d 4) or delayed SCH (McArt and Neves, 2020). This suggests that the duration and timing of SCH are critical factors influencing lactation performance and overall health in multiparous dairy cows. Although our initial hypothesis predicted that prepartum SCH, induced by EGTA and without acidifying the prepartum diet, would better prepare cows for the subsequent lactation, resulting in increased blood Ca concentrations, we did not observe a corresponding increase in early-lactation milk production. In fact, cows in the CON group produced more milk than cows in the EGTA group on certain days. However, this early lactation interaction did not affect total milk production over the entire lactation period. Our findings suggest that SCH in high-producing dairy cows may occur as a result of elevated milk production rather than a mechanism to increase production.

A study by Ben Chedly et al. (2010) reported that intramammary EGTA infusions resulted in decreased K^+ , Ca^{2+} , lactose, protein, and casein concentrations, alongside increased Na^+ and serum albumin concentrations in milk. Additionally, the authors observed an increase in blood lactose concentrations (Ben Chedly et al., 2010). These changes suggest compromised mammary epithelium integrity, leading to reduced milk yield (Ben Chedly et al., 2010). It is possible the decline in milk production could partially be attributed to decreased milk lactose content, given its osmotic properties (Ben Chedly et al., 2010). However, our study did not find significant differences in milk Ca^{2+} and lactose concentrations, suggesting the decrease in milk production was not due to lactose osmotic capability. However, the authors also reported increased MG apoptosis in EGTA-

infused glands, likely due to epithelial disruption (Ben Chedly et al., 2010). It is possible that the EGTA treatment in our study could be affecting the MG directly. Future research should explore this possibility.

CONCLUSIONS

Prepartum EGTA infusion effectively prevented SCH postpartum. However, the exact mechanism underlying the observed increase in blood Ca concentration remains unclear. Although early-lactation milk production was affected, there was no overall impact on total milk yield or DMI across the lactation period. Our results suggest that chelation of Ca close to parturition improves postpartum Ca concentrations by stimulating increases in PTH and serotonin in the circulation. Finally, pregnant cows require significantly more EGTA to induce SCH hypocalcemia. Further work should be performed to understand the mechanism regulating the impact of prepartum Ca manipulation on regulation of Ca metabolism postpartum.

NOTES

This research was supported by USDA National Institute of Food and Agriculture (NIFA; Washington, DC) funding 2020-67015-31260 and USDA Hatch formula fund 101955. Special thanks go to the USDA Dairy Forage Research Center (Prairie Du Sac, WI) staff for their help in conducting this experiment. All procedures involving animals were previously approved by the Institutional Animal Care and Use Committee of the University of Wisconsin–Madison under protocol number A006319-A03. Laura L. Hernandez is the incoming editor-in-chief for the *Journal of Dairy Science*. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: $1,25(OH)_2D_3$ = active form of vitamin D; 5-HT = serotonin; 5HTP = 5-hydroxy-L-tryptophan; AD-ICP = acid detergent insoluble protein; Adj CP = adjusted crude protein; aNDF = acid neutral detergent fiber; aNDFom = amylase-treated neutral detergent fiber on an organic matter basis; BE = base excess; CH = clinical hypocalcemia; CON = 6 h continuous intravenous infusion of saline solution; cwt = hundred weight; EE = ether extract; EGTA = egtazic acid; ESC = ethanol soluble carbohydrates; Hb = hemoglobin; iCa = ionized Ca; IR = infusion rate; Lact = lactation; ME_higgs (metabolizable energy) in the Cornell Net Carbohydrate and Protein System; MG = mammary gland; NDFD240 = undigested neutral detergent fiber at 240 hours; NDFom = neutral detergent fiber on an organic matter basis; OARDC = Ohio Agricultural Research and Development Center; pCO_2 = partial pressure of

CO₂; pO₂ = partial pressure of O₂; Protein Sol. = soluble protein; PTH = parathyroid hormone; PTHLH = parathyroid hormone-related protein; RDM = Redmond real salt/clay); SCH = subclinical hypocalcemia; tCa = total Ca; tCO₂ = total dissolved CO₂; TDN = total digestible nutrients; TFA = *trans* fatty acids; Trt = treatment; uNDFom = undigested neutral detergent fiber on an organic matter basis; WSC = water soluble carbohydrates.

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